

TorchJD: Jacobian Descent in PyTorch

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github.com/SimplexLab/TorchJD

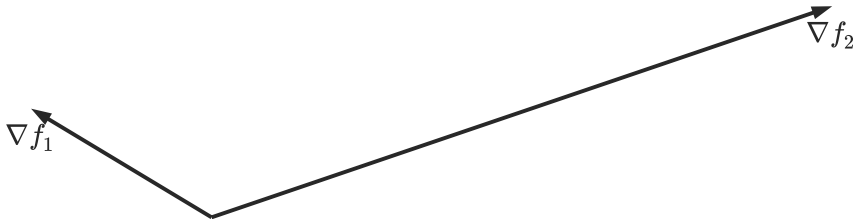


torchjd.org

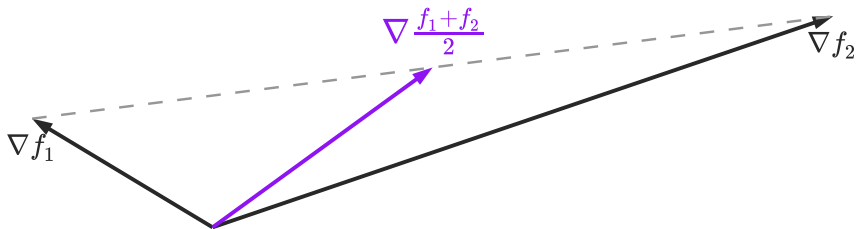


arxiv.org/pdf/2406.16232

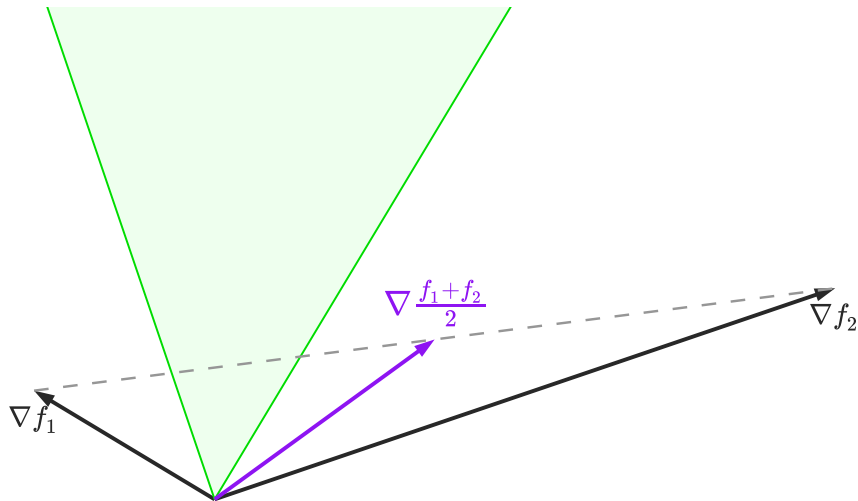
Jacobian descent intuition



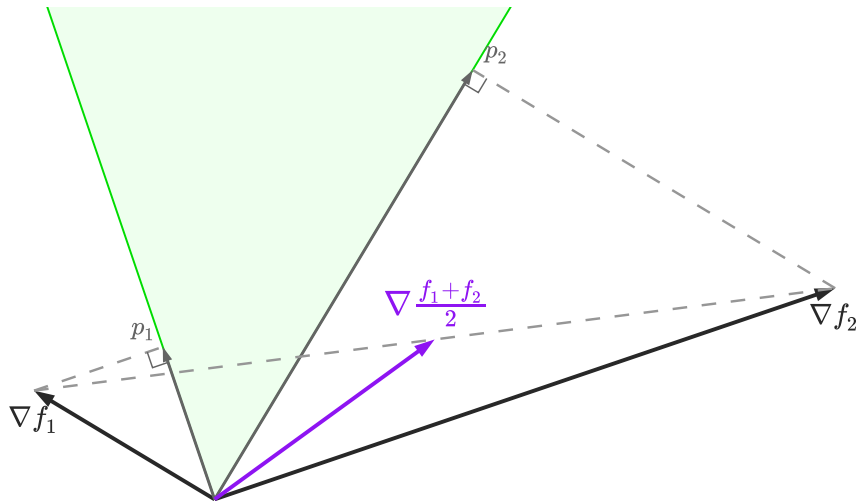
Jacobian descent intuition



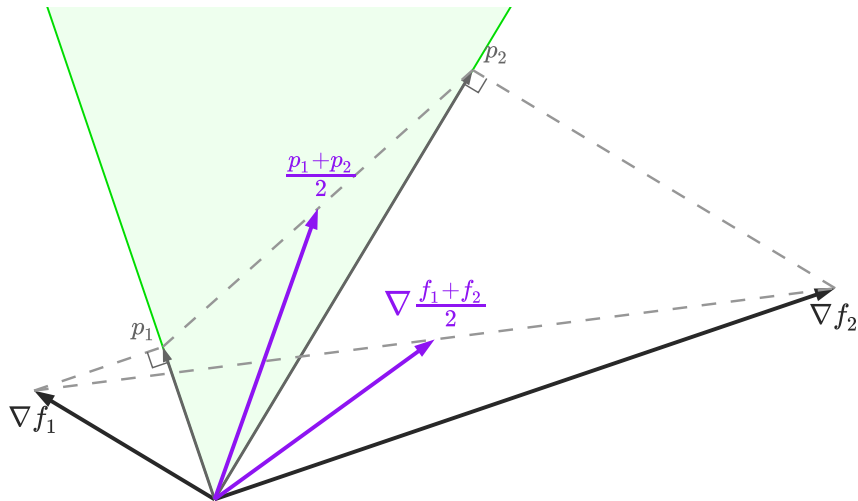
Jacobian descent intuition



Jacobian descent intuition



Jacobian descent intuition



TorchJD: core functions

Library	Function	Description
torch	autograd.grad	compute gradients
torch	autograd.backward	compute gradients and accumulate in .grad
torchjd	autojac.jac	compute jacobians
torchjd	autojac.backward	compute jacobians and accumulate in .jac
torchjd	autojac.jac_to_grad	aggregate .jac fields into .grad

TorchJD: basic usage

```
from torchjd import autojac
from torchjd.aggregation import UPGrad

batch_size = 16
loss_fn = MSELoss(reduction="none")

for x, y in dataloader:
    z = model(x) # shape: [16]
    losses = loss_fn(z, y) # shape: [16]
    autojac.backward(losses)
    # parameters now have a .jac field
    autojac.jac_to_grad(model.parameters(), UPGrad())
    # parameters now have a .grad field
    optimizer.step()
    optimizer.zero_grad()
```

More examples at torchjd.org.

You can use TorchJD to compute and aggregate Jacobians.

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Why should you care?

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Why should you care?

We need your help.